

**Exercise 1: (Lecture)**

Consider a matrix **A** of size  $n*m$  (where  $n \leq 20$ ,  $m \leq 30$ ). Write an algorithm that:

1. Reads (fills) the values of matrix **A** using two different methods (by row and by column).
2. Displays the maximum and the average of all elements in **A**.
3. Constructs matrix **B**, which represents the transpose of matrix **A**.

**Exercise 2: (Lecture)**

Given a **square matrix** of integers **A** with a size of  $n$  ( $n \leq 20$ ), write an algorithm that:

1. Reads (fills) the values of matrix **A**.
2. Constructs **matrix B** as follows: the matrix contains the **values of matrix A**, then we put the **value of the row index** on the elements of the main diagonal, and finally, the **value of the column index** on the elements of the secondary diagonal.
3. Calculates **SP**: the sum of the elements of the main diagonal of **B**, and **PS**: the product of the elements of the secondary diagonal of **B**.
4. Displays matrix **A** column by column, matrix **B** row by row, and the values of **SP** and **PS**.

**Exercise 3: (Lecture)**

Consider **A**( $n*m$ ) to be a matrix ( $n \leq 20$ ,  $m \leq 30$ ) of integers, where each row of the matrix contains elements **sorted in ascending** order.

Write an algorithm that inserts a given value **val** into each row of the matrix if it doesn't already exist. The insertion should preserve the ascending order of the elements. In the case of insertion, the last value of the row will be lost.

**Exercise 4: (Lecture)**

A **stochastic matrix** is a square matrix where each element is a real number between 0 and 1, and the sum of the elements in each row is equal to 1.

Write an algorithm that reads a square matrix **A** of order **N** ( $N \leq 20$ ), then checks if it is stochastic.

**Example:** This matrix is a **stochastic** matrix.

|     |     |     |
|-----|-----|-----|
| 0.5 | 0.3 | 0.2 |
| 0.2 | 0.8 | 0   |
| 0.3 | 0.3 | 0.4 |

**Exercise 5: (Lecture)**

Given a square matrix of integers **M** with a size of  $n*n$ , where  $n \leq 20$ . Write an algorithm that allows you to:

1. Read this matrix.
2. Calculate, for each column of this matrix, **the sum** of all elements that are **multiples of 3**, and place it on the main diagonal of **M**, as shown in the following example:

|   |   |    |   |    |   |   |   |    |   |    |  |                            |                    |
|---|---|----|---|----|---|---|---|----|---|----|--|----------------------------|--------------------|
| M | 1 | 18 | 5 | 45 | → | M | 9 | 18 | 5 | 45 |  | General term of an element |                    |
|   | 3 | 6  | 1 | 12 |   |   | 3 | 27 | 1 | 12 |  | Main Diagonal              | Secondary diagonal |
|   | 4 | 3  | 2 | 3  |   |   | 4 | 3  | 0 | 3  |  | By Row                     | $M[i, n-i-1]$      |
|   | 6 | 2  | 4 | 3  |   |   | 6 | 2  | 4 | 63 |  | By Column                  | $M[n-j-1, j]$      |

3. Display the resulting matrix.

**Exercise 6: (Lecture)**

Write an algorithm that reads the elements of a square matrix **A** of size  $n*n$  ( $n \leq 20$ ), then transposes it in-place without using another array, and finally displays the resulting matrix

**Exercise 7: (Lecture)**

Given a matrix of integers **M** containing **n** rows and **m** columns (**n**≤10, **m**≤20).

Write an algorithm that calculates and displays **the sum of all neighbouring** elements of an element **M [p, q]** (an element located in the **p<sup>th</sup>** row and **q<sup>th</sup>** column of **M**).

**Example** :the sum of neighbours of **M [2,3]** is:

$$1-5+2+6+12+3+0-13=6.$$

$$\begin{pmatrix} 4 & 1 & -5 & 2 \\ 7 & -13 & -1 & 6 \\ 8 & 0 & 3 & 12 \end{pmatrix}$$

**Exercise 8**

Write an algorithm that reads the dimensions **n** and **m** of a **matrix A** and verifies that they are less than or equal to 10; if they are not, it will display: 'The number of rows or columns is incorrect and exit the algorithm. In the favourable case, the algorithm:

1. Fills the matrix with values entered via the keyboard and displays **A**.
2. Displays the **sum, product, average** of all elements, and the minimum.
3. Stores the sum of each row in a slot of a vector **W**, then displays **W**.
4. Transfers the matrix into a vector **V** and then displays **V**.

**Exercise 9**

Given a matrix of real numbers **M** with a size of (**n\*m**), where **n**≤50 and **m**≤30. Write an algorithm that:

1. Reads this matrix.
2. Calculates and displays the number of elements strictly greater than a given real value **val** for all columns with even indices. **Example:** **n**=3, **m**=4, **val**=2.5

$$M = \begin{pmatrix} 1.2 & 0.4 & 2.8 & 7.3 \\ 2.5 & 30.2 & 4.4 & 1.0 \\ 3 & 1.5 & 6.8 & 3.2 \end{pmatrix} \quad \text{The number of elements strictly greater than 2.5 = 3.}$$

3. Builds and then displays a vector **V** of size **m**, such that the **j<sup>th</sup>** element contains the sum of the elements in the **j<sup>th</sup>** column of **M** if **j** is a multiple of 3; otherwise, it will contain the product of the elements in the **j<sup>th</sup>** column of **M**. **Example:**

$$M = \begin{pmatrix} 1.2 & 0.4 & 2.8 & 7.3 \\ 2.5 & 30.2 & 4.4 & 1.0 \\ 3 & 1.5 & 6.8 & 3.2 \end{pmatrix} \Rightarrow V = \begin{pmatrix} 1.2 * 2.5 * 3.6 \\ 0.4 * 30.2 * 1.5 \\ 2.8 + 4.4 + 6.8 \\ 7.3 * 1.0 * 3.2 \end{pmatrix} \Rightarrow V = \begin{pmatrix} 10.8 \\ 18.2 \\ 14 \\ 23.36 \end{pmatrix}$$

**Exercise 10**

Write an algorithm that performs **right rotations on each row** of a given **matrix A** of integers, with **n** rows (**n** ≤ 20) and **m** columns (**m** ≤ 30), such that the 1<sup>st</sup> row undergoes one rotation, the 2<sup>nd</sup> row undergoes two rotations, ..., and the **n**th row undergoes **n** rotations.

Very important:

- You must use only a single matrix (the matrix **A** that needs to be modified),
- Use variables to store the size of the matrix and indices to traverse the matrix,
- Use only a single additional variable for the rotations (nothing more).

**Example:** For the given matrix **A**

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{pmatrix} \Rightarrow \begin{pmatrix} 4 & 1 & 2 & 3 \\ 7 & 8 & 5 & 6 \\ 10 & 11 & 12 & 9 \end{pmatrix}$$